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# MTR KV

## Motion to Residency Reconciliation

Final Project Report

Submitted for

**Artificial Intelligence**

**BCSE306L**


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# Acknowledgement and Abstract

## Acknowledgement

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## Abstract

MTR-KV investigates how compressed video motion information and model feedback can guide the residency of visual key and value tensors. The prototype reconciles codec blocks with visual token geometry, tracks temporal states, emits generation tagged allocation descriptors, and commits page mappings through a software memory controller. A semantic correction path can promote important pages that appear quiet according to codec motion alone.

The implementation includes a deterministic simulator, real H.264 motion vector ingestion, a local demonstration dashboard, and an experimental ViT-Tiny path in which the final attention block reads tiered K/V tensors. Exact input equality permits bounded reuse across consecutive processed frames. T0 stores fp16 tensors on the selected device, T1 uses CPU int8 representations, and T2 stores compressed fp16 bytes.

Repository measurements show a median mean frame time of 55.570 ms for exhaustive reconciliation and 26.484 ms for the indexed path over three paired CPU runs, a 52.34 percent reduction in that comparison. A generated 216 frame demonstration records 862 page reuses and agreement with standard ViT on all compared frames. A separate natural scene run records no reuse and agreement on 170 of 171 frames. These observations establish a functioning software prototype and useful correctness evidence; they do not establish a production GPU speedup or downstream VLM accuracy.

Keywords: artificial intelligence, vision transformer, motion vectors, K/V tensors, memory residency, temporal reasoning, semantic feedback.

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## Figures and tables

Figures 1 to 5 describe the architecture, spatial mapping, temporal states, controller lifecycle, and reuse decisions. Figures 6 to 9 present simulated savings, paired timings, reuse counts, and classifier agreement. Tables summarize requirements, configurations, implementation modules, numerical results, test coverage, and future evaluation priorities.

# 1 Introduction and Problem Definition

## 1.1 Project context

Transformer based vision systems represent images as token sequences and form query, key, and value tensors during attention. Video introduces repeated visual content, yet the relevance and stability of individual regions can change over time. Treating every visual page identically can overlook opportunities to use a more economical representation for quiet content while preserving ready access to active or important regions.

MTR-KV studies a control path that connects compressed video observations to page residency decisions. Codec motion vectors are an accessible signal about inter frame prediction. They must be interpreted cautiously: low codec motion is neither proof of semantic irrelevance nor proof that a transformer feature has remained unchanged. The project therefore combines geometry reconciliation, temporal reasoning, semantic correction, and explicit generation checks. [1, 3]

## 1.2 Problem statement

The problem is to assign logical visual K/V pages to suitable storage tiers using decoder motion observations and model feedback, while ensuring that each consumer reads the correct ready generation. The design must reconcile different spatial grids, tolerate missing metadata, prevent noisy tier switching, and handle delayed updates without exposing stale page mappings.

## 1.3 Why the problem is relevant to AI

The project connects neural inference with a rule based reasoning layer. Attention provides a model derived signal; finite state temporal reasoning distinguishes persistent quiet behavior from a single low motion observation; and conservative rules determine when uncertain evidence should prevent aggressive placement. These mechanisms make the system suitable for studying the interaction between AI model signals and inference infrastructure.

## 1.4 Scope of the completed work

The completed artifact is a Python simulator and experimental CPU model integration. It includes real encoded video input, actual tensor representation operations, a browser dashboard, and testable control invariants. The live model is an image classifier, and only its final attention block uses the tiered K/V path. A full vision language application, GPU memory controller, and RTL implementation remain outside the demonstrated scope. [3, 4]

# 2 AI Foundations and Related Concepts

## 2.1 Attention and visual tokens

The Transformer introduced an architecture based on attention rather than recurrence or convolution for sequence modeling. The Vision Transformer subsequently demonstrated image classification using sequences of image patches. These works motivate the token and attention abstractions used in this project. MTR-KV changes how selected intermediate tensors are represented and accessed; it does not train a new recognition model. [1, 2]

In scaled dot product attention, queries select information from values according to their compatibility with keys. For a token feature matrix X, learned linear projections produce Q, K, and V. The attention operation can be expressed as follows:

Attention(Q, K, V) = softmax(Q K transpose / sqrt(d)) V

Here d is the key dimension. The storage experiment concerns K and V for visual patch tokens in one final ViT block. Prefix token K/V remains inside the model, and query projection still runs for every token. This is distinct from implementing a full autoregressive language model cache. [5]

## 2.2 Motion evidence and semantic evidence

Codec vectors describe how the decoder predicts blocks from reference frames. They operate on a different grid from neural patches and do not directly indicate object importance. Semantic feedback in the prototype aggregates class token attention into page scores. Such scores are model signals rather than ground truth explanations. A static but important region can therefore trigger a semantic promotion even when its motion activity is low. [3, 4]

## 2.3 Temporal reasoning and representation tradeoffs

A finite state engine uses confidence, activity thresholds, and persistence counters to avoid reacting independently to each frame. The controller then chooses among fp16, reduced precision, and compressed representations. These representations have different payload sizes and computational costs; reducing payload size can introduce quantization error or compression overhead. The report evaluates these effects through actual traces and baseline comparisons rather than assuming that compression implies faster inference.

# 3 Objectives and System Requirements

The primary objective is to implement an observable and testable path from codec motion observations to generation safe visual K/V residency. Secondary objectives are to integrate model derived feedback, demonstrate actual tensor consumption, and quantify the behavior under controlled and natural video inputs.

| Requirement | Implemented behavior |
|---|---|
| Motion ingestion | Extract exported motion vectors from compressed video through PyAV and FFmpeg. |
| Geometry correctness | Map model patch regions back through resize and crop transforms into source coordinates. |
| Temporal consistency | Maintain ACTIVE, STABLE, STATIC, TRANSITIONING, and UNKNOWN states. |
| Generation safety | Validate stream, page, generation, and readiness before mapping or reading. |
| Semantic correction | Promote low motion pages with high model derived importance through the controller. |
| Tensor consumption | Read controller selected patch K/V inside the final ViT attention block. |
| Auditability | Export JSONL decisions, configuration, timings, and classifier comparisons. |
| Demonstration | Provide synchronized decoded video and committed logical page mappings. |

## 3.1 Nonfunctional requirements

Deterministic scenarios and a fixed seed support repeatable simulator checks. Conservative fallback maintains an explicit uncertainty state when metadata is missing. The indexed reconciliation path must preserve the exhaustive path's decisions. The local dashboard must expose measured state clearly, and configurations must preserve the distinction between reference simulator geometry and the live model geometry. [3, 6]

## 3.2 Evaluation questions

The experiments ask whether semantic correction changes codec only placement, whether indexing preserves controller decisions while reducing reconciliation cost, whether exact reuse occurs on repeated content, and whether classifier outputs remain consistent with standard attention on the tested inputs. None of these questions alone establishes general task accuracy or hardware acceleration.

# 4 System Architecture

The architecture separates observations, reasoning, control, and tensor representation. Motion and geometry produce page activity; temporal state determines whether quiet behavior is persistent; descriptors request changes; and the controller publishes committed mappings. Semantic feedback enters through the same controller so that corrections follow the existing safety contract. [3, 4]

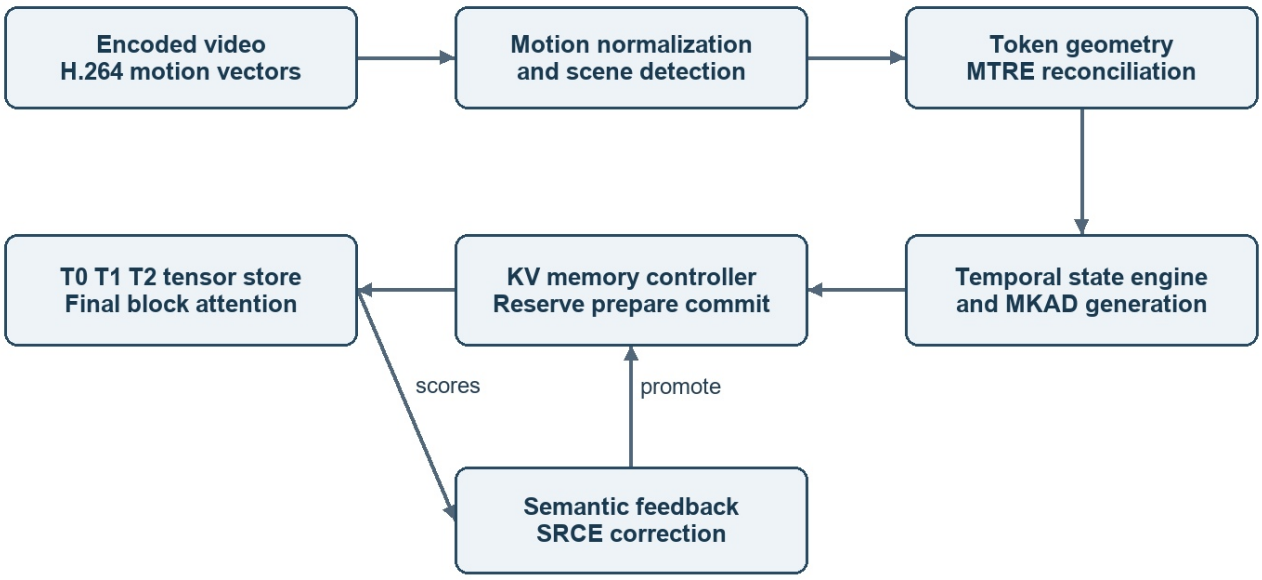



Figure 1. Implemented software architecture. The feedback edge promotes or corrects page placement after model observations. Diagram derived from the source module interfaces.

## 4.1 Main processing stages

The decoder supplies frame type, dimensions, timestamps, RGB frames, and exported motion vectors. Motion normalization bounds raw observations. MTRE maps the observations to patch tokens and logical pages. The temporal engine tracks state. MKAD descriptors carry requested allocations or migrations. KMC validates and commits them before consumers use the page.

## 4.2 Data and feedback boundaries

The simulator models capacity and event behavior, whereas the optional tensor store measures actual tensor payload operations. The live path uses one block of ViT-Tiny instead of the larger frozen reference geometry. The dashboard shows logical page identifiers rather than rectangular image regions. Separating these views prevents modeled occupancy, actual payload bytes, and displayed tiles from being mistaken for the same quantity.

# 5 Spatial Reconciliation

The Motion to Token Reconciliation Engine addresses the mismatch between codec blocks and model patches. A block can overlap several tokens, and a token can receive evidence from several blocks. A correct implementation must account for source dimensions, preprocessing scale, crop offsets, clipping, and invalid or padded regions. [3, 6]

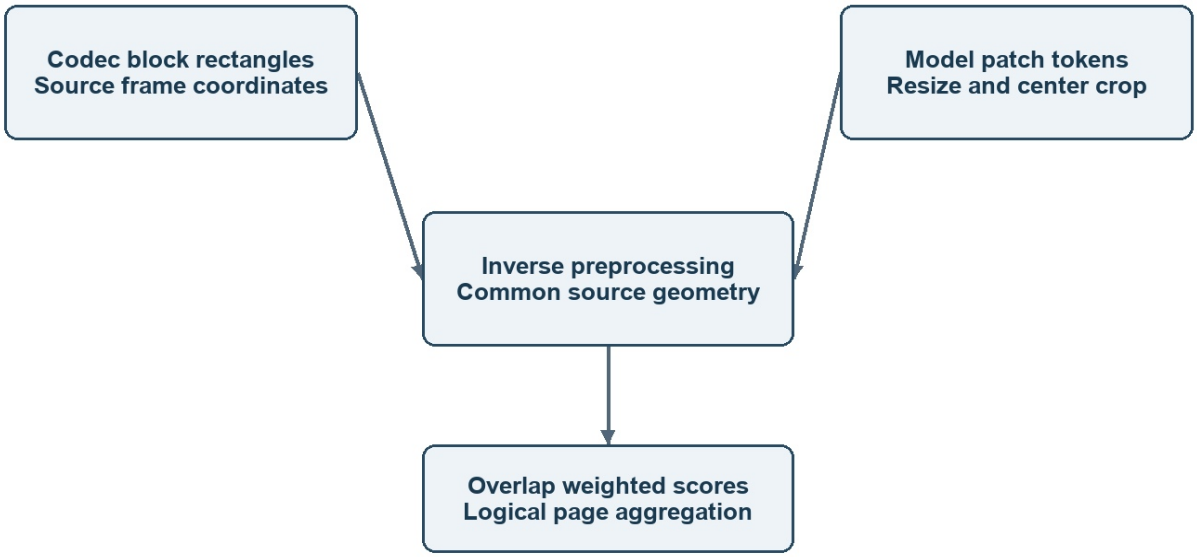



Figure 2. Spatial reconciliation in a shared source coordinate system. Token footprints are inverse mapped before intersections are evaluated.

## 5.1 Mapping and overlap

For model coordinates (x, y), scale (sx, sy), and source crop offset (cx, cy), the inverse mapping uses xsource = x / sx + cx and ysource = y / sy + cy. Token width and height are transformed in the same way. Rectangles are clipped to the source bounds. Intersection areas weight the contribution of overlapping codec blocks. Mixed active and quiet evidence is handled conservatively to avoid unsafe demotion.

## 5.2 Indexed and exhaustive paths

The exhaustive path checks candidate intersections directly and serves as a correctness oracle. The optimized path uses a tiled spatial index to reduce candidate work while preserving accumulation order and page aggregation. Randomized tests compare complete token and page results, including duplicate observations and empty inputs. The paired timing study changes the reconciliation path while holding the other principal settings constant. [6]

# 6 Temporal Reasoning and Activity Scoring

Motion normalization uses bounded Manhattan magnitude: M = min((abs(dx) + abs(dy)) / 32, 1). Residual energy, when available, is normalized separately. Confidence remains explicit so unreliable evidence can trigger UNKNOWN behavior instead of being interpreted as still content. [3]

The default full activity score is A = clip(0.50 M + 0.25 R + 0.10 P + 0.15 C, 0, 1), where R is residual, P is the persistence feature, and C is confidence. The compressed video adapter does not supply measured residual energy; the live evaluation must therefore record that limitation.

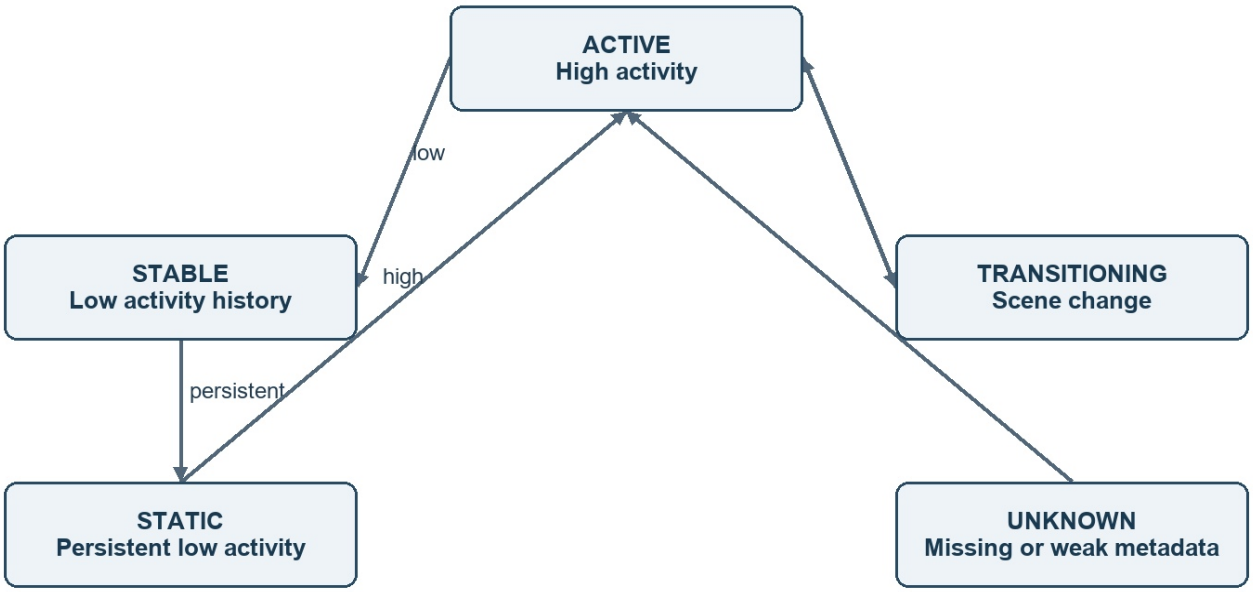



Figure 3. Temporal reasoning overview. Invalid or low confidence metadata can enter UNKNOWN from any state; valid recovery reevaluates the current observations. The diagram summarizes the code rather than enumerating every guard.

## 6.1 Hysteresis and persistence

The low threshold is 0.20 and the high threshold is 0.35. A reliable high activity observation returns the page to ACTIVE. Low activity starts or extends a quiet history, and persistent evidence can reach STATIC after the configured progression. The reference persistence and stable frame parameters are 4 and 6. Intermediate activity preserves low path progress rather than forcing a rapid oscillation between extremes.

Scene change evidence enters TRANSITIONING, while intra frames reset temporal assumptions and broaden residency. Missing vectors are never silently treated as zero motion. These rules prioritize conservative correctness over a high reported savings rate.

# 7 Generation Safe Controller Operations

Asynchronous observations create a correctness problem: an older descriptor or delayed semantic update can arrive after a newer frame has become current. MTR-KV tags operations with stream, page, and generation identity and uses explicit readiness. Consumers must not read a partially prepared representation or a stale generation. [3]

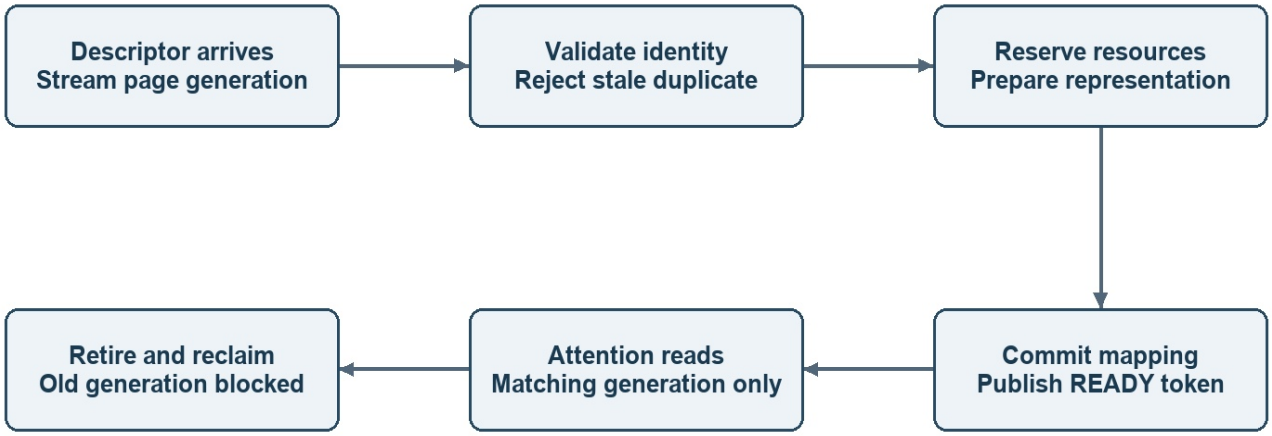



Figure 4. Controller transaction and consumption lifecycle. Rejected or stale operations do not publish a usable new mapping.

## 7.1 Descriptor and commit contract

Motion informed K/V allocation descriptors request operations such as allocation, promotion, demotion, and prefetch. The controller checks identity, coalesces appropriate queued operations, reserves resources, prepares the representation, and commits a mapping. A ready token identifies a generation that can be consumed. The model must wait for the matching mapping before attaching or reading tensors.

## 7.2 Fault handling

The apply status can distinguish APPLIED, STALE_DROP, DUPLICATE, RETRY, FALLBACK, and ABORT. Missing metadata, queue pressure, scene transitions, allocation failures, and late semantic evidence have explicit handling paths. Resource limits and an emergency pool support conservative promotions, while stale reads remain invalid.

## 7.3 Safety invariant

A consumer can use a page only when the requested generation agrees with a committed ready mapping. Reuse does not waive this requirement: retained content is rebound only after the next generation completes its normal control path. This separates the age of tensor content from the identity of the current mapping.

# 8 Semantic Feedback and Tensor Storage

## 8.1 Semantic residency correction

The semantic correction path addresses a weakness of motion only placement. A low motion page may still influence the model's decision. The live adapter aggregates final block class token attention into page scores and submits them as generation tagged semantic updates. Default low codec and high semantic thresholds are 0.20 and 0.70. Corrections pass through KMC; they do not directly overwrite a page table. [3, 4]

| Tier | Live tensor representation | Interpretation |
|---|---|---|
| T0 | fp16 on the selected torch device | Preferred representation for hot pages. |
| T1 | Symmetric int8 on CPU | Reduced precision with quantization overhead and possible error. |
| T2 | Compressed contiguous fp16 bytes on CPU | Compression preserves the stored fp16 representation. |
| T3 | Optional simulator reconstruction state | Disabled in the default configuration; not a live tensor tier. |

## 8.2 Model consumption path

Without the inference flag, captured tensors can be stored after ordinary attention, which demonstrates representation operations but not consumption by attention. With --kv-inference, the final attention block writes patch K/V to the controller selected tiers and reads them back within attention. Prefix tokens remain in the model. The baseline comparison runs standard ViT on the same input and reports top 1 agreement and mean absolute logit error.

## 8.3 Accounting boundaries

The capacity model includes configured geometry, page padding, and tier ratios. Actual tensor payloads measure bytes in the implemented representations. Reuse validation inputs introduce another allocation that must be reported separately. Peak payload is therefore not equivalent to total process memory or GPU allocator usage. Compression time, data conversion, and equality checks must be considered alongside byte counts.

# 9 Cross Frame Reuse

The optional reuse extension retains K/V only for unchanged patch inputs in one final ViT block. Low motion alone does not authorize reuse. Earlier attention layers can change the input features of a visually quiet region when another region changes, so the adapter verifies exact page input equality before skipping K/V projection. [5]

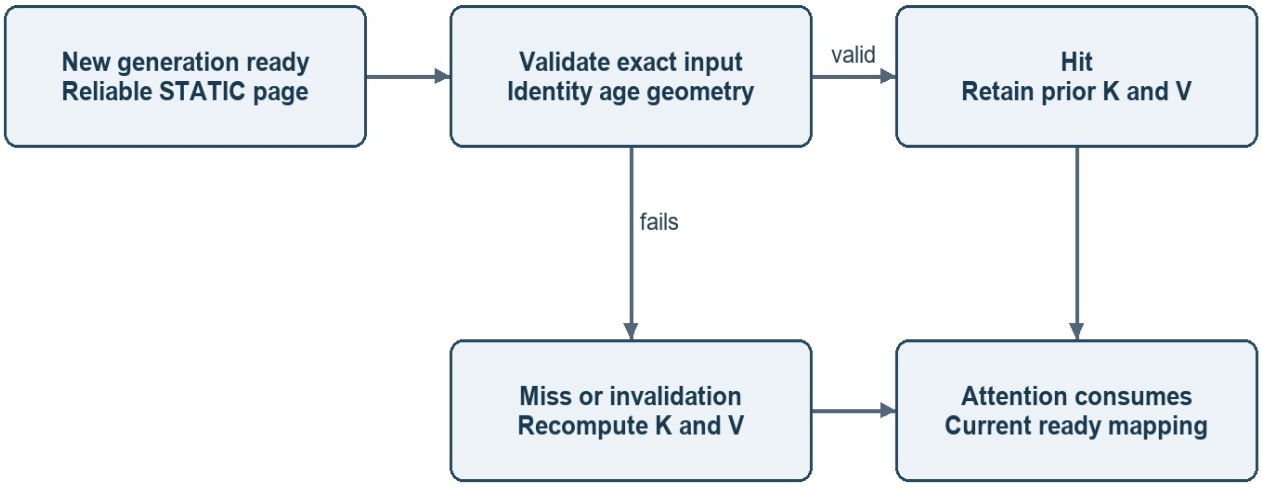



Figure 5. Reuse eligibility and refresh path. Every hit still requires a new committed ready mapping and a valid content age.

## 9.1 Reuse eligibility

A candidate needs reliable metadata, a STATIC page, the immediately preceding generation, matching controller identity, compatible geometry and representation settings, and content within the configured origin age window. The default window is eight processed generations. Changed features, scene transitions, missing metadata, skipped generations, parameter changes, failed inference, or expired content force refresh.

## 9.2 Work retained and work skipped

Query projection runs for all tokens. Prefix token and missed patch K/V are projected normally. Hits read previously stored payload through the current ready mapping; a tier change migrates its representation. Earlier ViT blocks still execute, and validation input retention, cloning, equality checks, compression, and split projection can add cost. Zero hits on moving natural content are a valid result, not an implementation failure.

# 10 Implementation and Demonstration

| Module or interface | Responsibility |
|---|---|
| motion.py and mtre.py | Normalize codec evidence and reconcile source geometry. |
| temporal.py and predictor.py | Track temporal state and scene changes. |
| mkad.py and controller.py | Build operations and enforce control sequencing. |
| memory.py and tensor_store.py | Model occupancy and store real tensor representations. |
| video.py and vision_feedback.py | Decode video and connect ViT feedback and K/V inference. |
| dashboard.py and frontend | Serve synchronized video, page mappings, and trace controls. |
| simulator.py and evaluate.py | Run scenarios and summarize exported measurements. |

## 10.1 Demonstration workflow

The demo command generates or reuses an 18 second H.264 clip with still content, a moving object, still content again, and a scene cut. The local dashboard pairs each decoded generation with the committed page mapping and measured vector counts. Restart replays the input; Stop ends the stream; Save trace exports measured image free JSONL records. [4]

The source may also be an FFmpeg compatible encoded file or stream. Android camera integration requires the documented USB and encoded camera setup. Generic browser uploads and raw webcam capture are not implemented. The default server binds locally, and model weights may need an initial download.

## 10.2 Reproducible commands

```
python -m mtr_kv run --scenario T03 --mode all
python -m mtr_kv report
python -m mtr_kv demo
python -m mtr_kv demo --semantic-model --kv-inference --kv-reuse --compare-baseline
python -m pytest -q
```

The repository's dependency file pins av 19.0.1, torch 2.14.1, timm 1.0.30, and torchvision 0.29.1, with NumPy and Pillow minimum versions. The recorded experiment runtime is macOS ARM64 with CPU tensor execution. [3, 9]

# 11 Experimental Methodology

## 11.1 Evidence groups

Four groups of evidence are kept separate: deterministic simulation, paired reconciliation timing, a generated video demonstration, and a natural scene check. The T03 simulator was rerun locally for this report. Model backed timings and test counts are retained repository experiment records rather than new measurements on the report authoring computer. [6-9]

| Setting | Reference simulator | Live model evaluation |
|---|---|---|
| Model input | 224 by 224 | 224 by 224 |
| Patch size | 14 pixels | 16 pixels |
| Modeled attention layers | 24 | 1 final block |
| K/V heads and head dimension | 16 and 128 | 3 and 64 |
| Page size | 32 tokens | 32 tokens |
| Tensor execution | Capacity and events | ViT-Tiny on CPU |
| Principal outputs | Pages and modeled bytes | Payload, timing, reuse, agreement |

## 11.2 Paired timing protocol

The primary manifest records three alternating oracle and indexed pairs with the same final code, input clip, crop, model, representations, and reuse configuration. Each run decodes 60 frames; the first six are retained but excluded from summary timing, leaving 54 evaluated frames. Decision hashes are compared between paths. The reported summary is the median of the three per run mean frame times. [6]

## 11.3 Metrics and interpretation

The report uses modeled bytes saved for simulation; mean and p95 frame time for video processing; reused and recomputed page counts for reuse; and agreement with standard ViT for classifier consistency. Frame timing includes the per frame baseline comparison but excludes initial model loading, dashboard rendering, and presentation pacing. Agreement has no ground truth labels and must not be called recognition accuracy.

# 12 Simulation Results

T03 exercises low codec activity with high semantic importance. The local rerun used the frozen reference configuration and seed 20260922. Baseline, codec only, and full correction modes were evaluated on the same deterministic scenario. The following values are cumulative simulator metrics, not physical memory measurements or a measured peak reduction. [7]

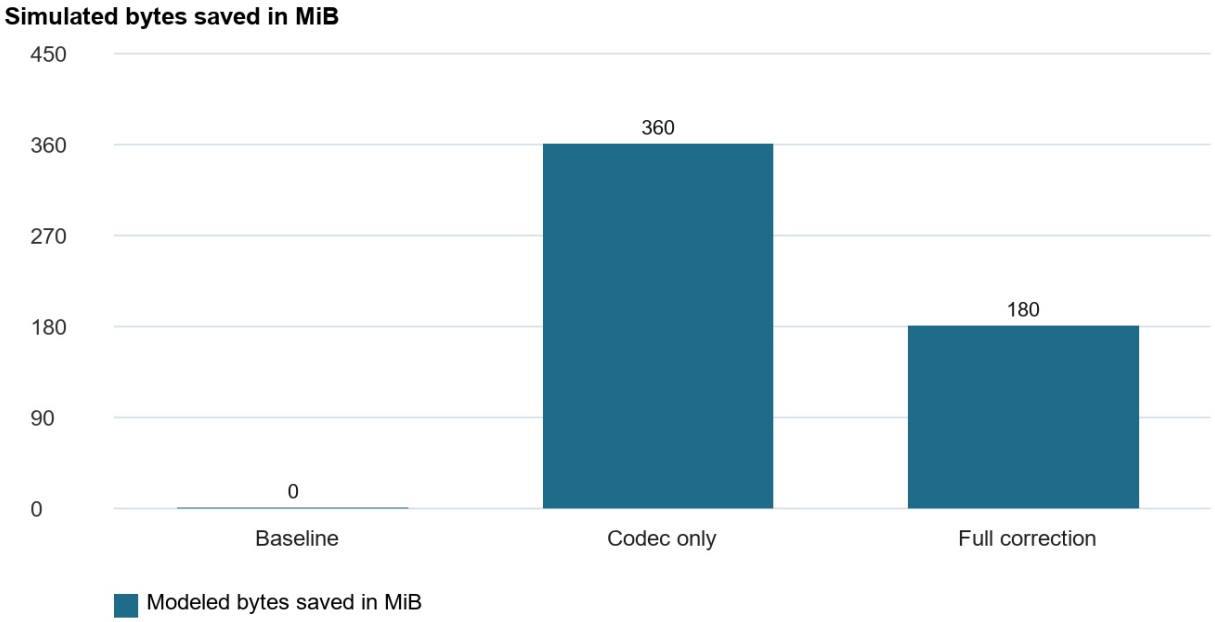



Figure 6. Modeled bytes saved for T03. MiB uses 1,048,576 bytes. Data: tmp/local-check/simulation.json and report.md.

| Mode | Pages avoided | Bytes saved | Rescues |
|---|---|---|---|
| Baseline | 0 | 0 | 0 |
| Codec only | 80 | 377,487,360 | 0 |
| Full correction | 40 | 188,743,680 | 40 |

## 12.1 Interpretation

Codec only mode reports greater savings because it does not restore the low motion pages that carry high semantic scores. Full correction rescues 40 pages and therefore retains more modeled storage. The lower savings in full mode reflect the intended protection of important content. All three runs record a 0 percent fallback rate and zero stale drops for this scenario; adversarial safety needs separate tests.

This result illustrates why minimizing stored pages is not the sole objective. A useful residency policy must preserve required information and control invariants while reducing avoidable resource use.

# 13 Paired Optimization Results

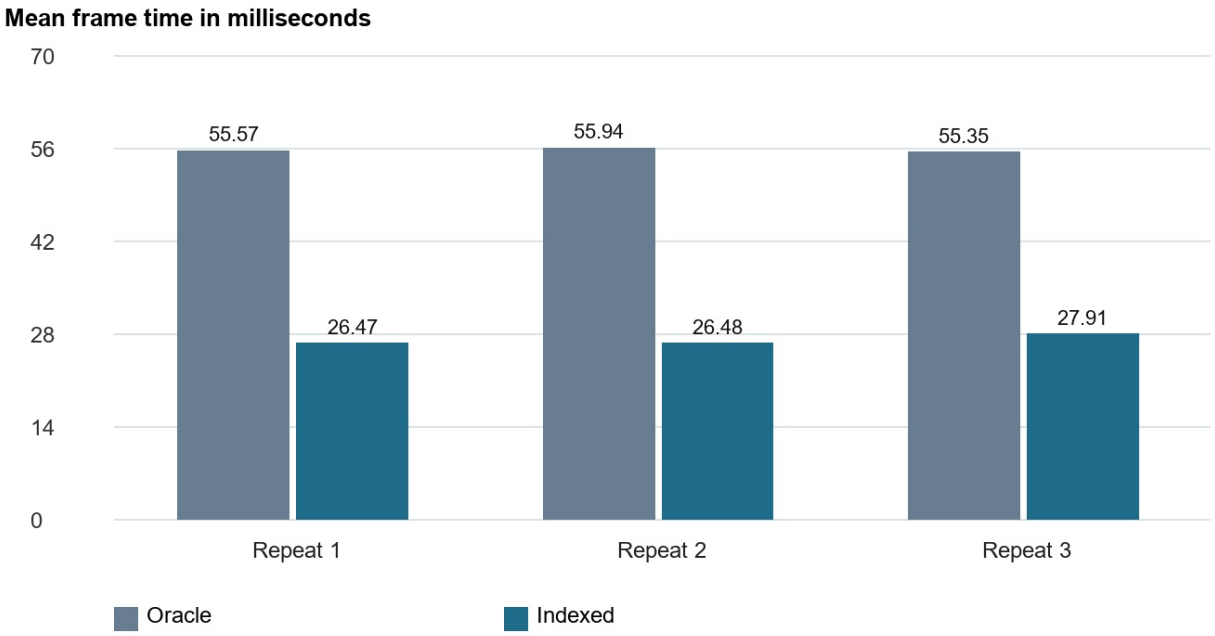



Figure 7. Mean frame time in three paired CPU runs. Each mean excludes six warmup frames and summarizes 54 frames. Source: paired-serial/manifest.json. [6]

| Repeat | Oracle mean ms | Indexed mean ms |
|---|---|---|
| 1 | 55.570 | 26.467 |
| 2 | 55.944 | 26.484 |
| 3 | 55.347 | 27.909 |

## 13.1 Quantitative finding

The median of per run means is 55.570 ms for the oracle and 26.484 ms for the indexed path. The observed ratio is 2.098. The relative reduction is (55.570 - 26.484) / 55.570 times 100 = 52.34 percent. Matching paired decision hashes support equivalent controller decisions for these recorded runs. [6]

## 13.2 What this comparison establishes

The result supports the indexed reconciliation optimization on the tested CPU workload. It compares two reconciliation implementations within the same experimental pipeline, which also performs a baseline inference comparison. It is not a comparison against standard ViT alone and does not establish a 2.098 times gain for a complete transformer application, a VLM, or a production GPU controller.

Three repeats provide useful repeatability evidence but do not characterize all machines or video content. A larger study should randomize run order, monitor system load and temperature, and preserve separate decoding, controller, inference, compression, and transfer measurements.

# 14 Video and Model Evaluation

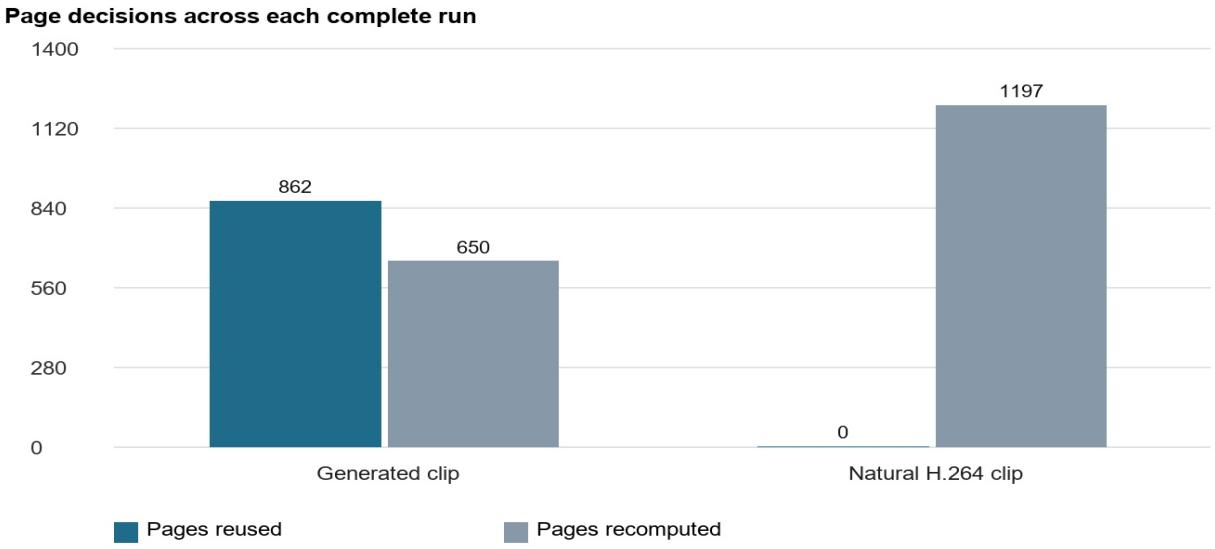



Figure 8. Actual page reuse and recomputation totals. The clips have different lengths and content, so totals describe each run rather than a controlled cross clip performance comparison. [8]

| Metric | Generated clip | Natural H.264 clip |
|---|---|---|
| Compared frames | 216 | 171 |
| Top 1 agreement | 216 / 216 | 170 / 171 |
| Pages reused | 862 | 0 |
| Pages recomputed | 650 | 1,197 |
| Mean frame time ms | 28.760 | 28.574 |
| p95 frame time ms | 38.854 | 34.261 |
| Mean absolute logit error | 0.000223 | 0.001712 |
| Metadata fallback rate | 1.85% | 11.11% |

Generated content produces a reuse fraction of 862 / (862 + 650) = 57.01 percent of page decisions. Natural content produces zero exact reuse. Both runs report a peak tensor payload of 150,528 bytes and a separate peak retained validation input size of 150,528 bytes. These peaks must not be presented as total application memory. The original VP8 natural source takes metadata fallback throughout; the table uses the derived H.264 clip. [8]

# 15 Testing and Verification

The repository retains a pytest record of 128 passed tests in 114.24 seconds. This is a prior recorded verification run, not a newly executed full suite on the report authoring machine. The test files cover geometry, temporal behavior, generation safety, races, fallback, tensor storage, reuse, decoding, dashboard behavior, and evaluation. [9]

| Area | Invariant or behavior checked |
|---|---|
| Geometry | Indexed and oracle token/page results agree; crop and overlap remain valid. |
| Temporal reasoning | Persistence and hysteresis limit unstable transitions. |
| Generation control | Stale mappings and late updates cannot authorize an old read. |
| Race handling | Adversarial ordering preserves the control contract. |
| Semantic correction | Important low motion pages can be rescued through KMC. |
| Tensor storage | T2 scalar, empty, and multidimensional round trips preserve stored fp16 content. |
| Reuse invalidation | Changed features, gaps, age expiry, and controller changes refresh tensors. |
| Integration | Video decoding, dashboard traces, and summary metrics follow their interfaces. |

## 15.1 Additional verification in this report

The T03 experiment and report generation completed successfully in the local working copy. The resulting values match the semantic rescue interpretation in Section 12. The source archive also preserves paired decision hashes, runtime information, JSONL traces, and evidence packages, making it possible to audit a numerical claim back to a particular experimental record.

## 15.2 Remaining validation

A fresh full environment installation and model backed rerun on the target demonstration computer is still needed before claiming that every path works there. Pretrained weight availability, device support, codec behavior, and operating system differences can affect the live demo. Neither a passing unit suite nor a favorable generated clip replaces representative workload evaluation.

# 16 Discussion and Limitations

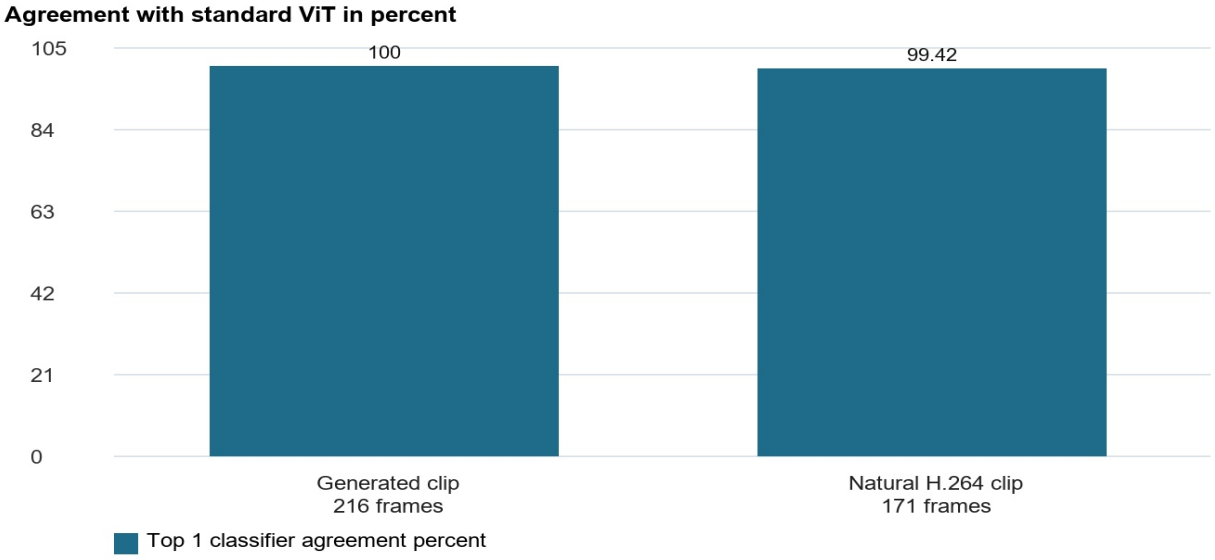



Figure 9. Agreement with standard ViT predictions. The vertical axis starts at zero. Agreement measures consistency between implementations, not labeled task accuracy. [8]

## 16.1 Relationship between efficiency and fidelity

T03 shows that semantic rescue can deliberately reduce the apparent storage benefit of a motion only policy. Natural video shows that exact reuse may disappear when features change continuously. These results support a cautious design: geometry, current generation identity, semantic importance, and feature equality all matter. A high reuse rate on a constructed still sequence cannot be generalized to arbitrary video.

## 16.2 Principal limitations

The model is ViT-Tiny, and only its final attention block consumes tiered patch K/V. Earlier blocks execute normally. T1 and T2 are CPU backed; this is not a measured GPU memory hierarchy. Codec residual energy is not available from the current adapter. Attention scores are not semantic ground truth. Quantization and split projections can change logits. Reuse validation consumes additional memory and compute. The evaluation includes a small number of clips and does not establish energy savings, broad throughput gains, labeled recognition quality, or production scalability.

The controller and diagrams describe a software embodiment. Hardware implementation, full VLM integration, patent novelty, and institutional filing status require separate evidence and are not conclusions of this academic evaluation.

# 17 Conclusion and Future Work

## 17.1 Conclusion

MTR-KV provides a working software path from encoded video motion observations to generation safe visual K/V residency. The project integrates spatial reconciliation, persistent temporal state, semantic correction, tiered tensor representations, and an observable local demonstration. Its model path reads actual controller selected patch K/V inside the final ViT attention block, while bounded exact reuse avoids selected K/V projections when validation succeeds.

The evidence supports three concrete conclusions. Semantic feedback changes motion only placement in the deterministic rescue scenario. Indexed reconciliation preserves recorded controller decisions while reducing the median mean frame time from 55.570 ms to 26.484 ms in the paired CPU experiment. Reuse is workload dependent: 862 pages are reused on the generated sequence, whereas the natural sequence records none. Classifier agreement is high on these inputs, but task accuracy and full application speedup remain unestablished.

## 17.2 Future work priorities

| Priority | Proposed extension | Evaluation needed |
|---|---|---|
| 1 | Reproduce on target hardware | Fresh environment, identical traces, repeated device timing. |
| 2 | Expand the video workload | Diverse motion, camera movement, scene cuts, and codec conditions. |
| 3 | Evaluate task quality | Labeled classification or VLM tasks with quality and calibration metrics. |
| 4 | Improve memory accounting | Total process memory, allocator peaks, validation costs, and transfer bytes. |
| 5 | Extend model integration | More blocks or a selected VLM with end to end correctness checks. |
| 6 | Study accelerator implementation | Controlled GPU memory placement and hardware feasibility experiments. |

The next stage should prioritize reproducible task level evaluation over broader performance claims. Any future relaxation of exact equality must define an error budget and test its effects on downstream quality before it is presented as safe reuse.

# 18 References

[1] A. Vaswani et al. Attention Is All You Need. Advances in Neural Information Processing Systems, 2017. arXiv:1706.03762. https://arxiv.org/abs/1706.03762

[2] A. Dosovitskiy et al. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. International Conference on Learning Representations, 2021. arXiv:2010.11929. https://arxiv.org/abs/2010.11929

[3] MTR-KV project repository. README.md; configs/reference.json; mtr_kv/config.py, motion.py, mtre.py, temporal.py, data_objects.py, controller.py, and tensor_store.py. Local project snapshot supplied for this report. Repository identifier: Shaurya002800/MTR-KV.

[4] MTR-KV project documentation. docs/DEMO.md and frontend/README.md. Demonstration workflow, interface behavior, and model consumption boundaries.

[5] MTR-KV project documentation. docs/KV_REUSE.md. Exact feature validation, generation rebinding, invalidation rules, and reuse limitations.

[6] MTR-KV optimization records, 21 October 2026. docs/OPTIMIZATION.md; results/optimization-20261021/paired-serial/manifest.json and repeat 1 to 3 oracle/indexed traces. Primary paired reconciliation comparison.

[7] Local deterministic T03 rerun, 21 October 2026. tmp/local-check/simulation.json, frames.csv, and report.md. Frozen reference configuration; seed 20260922. Values regenerated from the supplied implementation.

[8] MTR-KV video evaluation records. results/optimization-20261021/full-demo-serial-evaluation.json; results/freeze-v1/derived-h264-evaluation.json; original-vp8-evaluation.json. Generated and natural video measurements.

[9] MTR-KV verification and provenance records. results/optimization-20261021/test-results.txt and test-results.xml; output/evidence/runtime-and-provenance.json; tests/test_*.py. Recorded runtime and 128 test results.

## Report identification

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